Two-temperature cyclic compression for the Gyarmati-Hennecart-Ruzsa constant

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Status and result

This note gives a proof, with a Lean-verified finite instance in §6, that

$$\boxed{C_{3a} > 1.21418}.$$

The construction is not a retuning of the earlier six-generator mask. Its inner alphabet is the much smaller numerical-semigroup truncation

$$M = \langle 4, 5\rangle \cap [0, 80] = \{0, 4, 5, 8, 9, 10\} \cup [12, 80],$$

and its main new ingredient is a **two-temperature entropy theorem**: the difference family and the sum-family upper bound are evaluated at separate dual parameters. The old one-temperature theorem forced those two parameters to coincide.

At the exact rational parameters

$$x = \frac{8149}{10000}, \qquad y = \frac{42571}{50000}, \qquad R = \frac{249293}{10000},$$

the display-only fixed-rate value is

$$1 + \frac{\rho - \sigma}{\log R} = 1.2141808960977854942000570924 9 \ldots,$$

while the entropy-matched limiting value is

$$2 - \frac{\sigma}{\rho} = 1.2141811809505310474549692114 8 \ldots.$$

The conservative claim is only the strict rationally certified inequality $C_{3a} > 1.21418$.

The numerical inputs are checked by two independently structured programs. The strict finite bound is also proved end to end in Lean 4, as described in §6–7. The general limiting theorem remains a written argument. This package is not an externally refereed publication.

1. The finite-set principle

We use the following lemma from K. Gyarmati, F. Hennecart, and I. Z. Ruzsa, *Sums and differences of finite sets*, Functiones et Approximatio Commentarii Mathematici **37** (2007), 175-186, Section 2, pp. 178-179, DOI 10.7169/facm/1229618749.

GHR finite-set lemma. Let $U \subset \mathbb{N}_0$ be finite and nonempty, with $0 \in U$. Put

$$s = |U + U|, \qquad d = |U - U|, \qquad q_U = 2 \max U + 1.$$

If $d < q_U$, then

$$\theta(U) = 1 + \frac{\log(d/s)}{\log q_U}$$

is an admissible exponent for the GHR problem. More precisely, for every $K > 1$ there are arbitrarily large finite integer sets A, B such that $|A + B| \le K|A|$ and

$$|A - B| \ge \left(\frac{2(K-1)}{3K}\right)^{5/4} |A + B|^{\theta(U)}.$$

Thus a single finite U satisfying the strict difference-set condition supplies a lower bound for C_{3a}.

2. Universal cyclic compression

The outer compression step does not depend on the numerical semigroup.

Lemma 2.1 (cyclic quotient with controlled collisions)

Let $V \subset \mathbb{Z}^n$ be finite and contain zero. Let $W \subset V - V$ have $T = |W| \ge 2$, and suppose every coordinate of every word in W has absolute value at most B_0. If p is prime and

$$p > 2B_0, \qquad p > |V - V|,$$

then there is a linear map

$$\phi_c(z) = \sum_{i=1}^{n} c_i z_i \pmod p$$

such that, after multiplying ϕ_c by a nonzero scalar if necessary, the least-nonnegative representative set

$$A = \mathrm{rep}_{[0,p-1]}\, \phi_c(V)$$

satisfies

$$\max A = p - 1, \tag{2.1}$$

$$|A + A| \le 2|V + V|, \tag{2.2}$$

$$|A - A| > \frac{T}{2}, \tag{2.3}$$

and

$$|A - A| \le 2|V - V| - 1 < 2p - 1 = 2\max A + 1. \tag{2.4}$$

The coefficient vector can be chosen canonically by conditional expectations, followed by canonical rescaling.

Proof

Choose all coefficients $c_1, \dots, c_n$ independently and uniformly in $\mathbb{F}_p$. For distinct $w, w' \in W$, write $h = w - w'$. Some coordinate h_j is a nonzero integer with $|h_j| \le 2B_0 < p$, and hence is nonzero in $\mathbb{F}_p$. Holding the other coefficients fixed, exactly one value of c_j solves $\phi_c(h) = 0$. Therefore

$$\Pr(\phi_c(w) = \phi_c(w')) = \frac{1}{p}.$$

If $C(c)$ is the number of unordered colliding pairs in W, then

$$\mathbb{E}C(c) = \frac{\binom{T}{2}}{p} < \frac{T}{2},$$

because $T \le |V - V| < p$. Choose c with $C(c) < T/2$. This choice can be made coefficient by coefficient by minimizing the conditional expected final collision count and breaking ties by the least residue.

If a fiber of $\phi_c|_W$ has size t, it loses $t-1$ image points but contributes $\binom{t}{2} \ge t-1$ colliding pairs. Consequently

$$|\phi_c(W)| \ge T - C(c) > T/2.$$

Since $T \ge 2$, the image $\phi_c(V)$ is not $\{0\}$. Choose the least specified $v \in V$ with $\phi_c(v) \ne 0$, and multiply the map by $-\phi_c(v)^{-1}$. This does not change any cardinality and puts -1 in the image. Since zero is also in the image, the representative set contains $0, p-1$, proving (2.1).

Every residue has at most two representatives in $[0, 2p-2]$, so

$$|A + A| \le 2|\phi_c(V + V)| \le 2|V + V|.$$

Distinct residues in $\phi_c(W)$ give distinct integer differences, which proves (2.3). In $[-(p-1), p-1]$, residue zero has one representative and each nonzero residue has at most two. Hence

$$|A - A| \le 2|\phi_c(V - V)| - 1 \le 2|V - V| - 1 < 2p - 1.$$

This proves the lemma. □

3. The two-temperature theorem

Let $M \subset \mathbb{N}_0$ be finite, contain zero, and contain a positive integer. Define

$$S = M + M, \qquad D = M - M,$$

and, for $d \in D$,

$$\kappa(d) = \min\{a + b : a, b \in M,\ a - b = d\}.$$

For $x, y \in (0,1)$, put

$$P_-(x) = \sum_{d \in D} x^{\kappa(d)}, \qquad P_+(y) = \sum_{s \in S} y^s,$$

$$\mu = \frac{\sum_{d \in D} \kappa(d) x^{\kappa(d)}}{P_-(x)},$$

$$\rho = \log P_-(x) - \mu \log x, \qquad \sigma = \log P_+(y) - \mu \log y.$$

Theorem 3.1 (two-temperature cyclic quotient)

If $\sigma < \rho$, then

$$\boxed{C_{3a} \geq 2 - \frac{\sigma}{\rho}.} \tag{3.1}$$

More generally, for every real $r > \rho$,

$$\boxed{C_{3a} \geq 1 + \frac{\rho - \sigma}{r}.} \tag{3.2}$$

In particular, if a rational number $R > 1$ satisfies $\rho < \log R$, then

$$\boxed{C_{3a} \geq 1 + \frac{\rho - \sigma}{\log R}.} \tag{3.3}$$

Proof

Give D the symmetric Gibbs distribution

$$p_d = \frac{x^{\kappa(d)}}{P_-(x)}.$$

Its entropy is

$$H(p) = -\sum_d p_d \log p_d = \log P_-(x) - \mu \log x = \rho.$$

For every positive $d \in D$, set

$$n_d^{(n)} = n_{-d}^{(n)} = \lfloor n p_d \rfloor,$$

and put the unused count into zero:

$$n_0^{(n)} = n - 2 \sum_{d>0} \lfloor n p_d \rfloor.$$

These counts are nonnegative, sum to n, and their normalized frequencies converge to p. Moreover,

$$\sum_d n_d^{(n)} \kappa(d) = 2 \sum_{d>0} \lfloor np_d \rfloor \kappa(d) \le n\mu.$$

Let

$$L_n = \left\lfloor \frac{n\mu}{2} \right\rfloor,$$

$$V_n = \left\{ a \in M^n : \sum_i a_i \le L_n \right\},$$

and let $W_n \subset D^n$ be the type class with counts $n_d^{(n)}$.

Choose minimum-cost witnesses $\alpha_d, \beta_d \in M$ with

$$\alpha_d - \beta_d = d, \qquad \alpha_d + \beta_d = \kappa(d),$$

choosing the witnesses for $-d$ by swapping those for d. Every word in W_n has coordinate sum zero. Thus its total α-budget equals its total β-budget, while their sum is an even integer at most $n\mu$. Each budget is therefore at most L_n, and

$$W_n \subset V_n - V_n.$$

Writing $T_n = |W_n|$, Stirling's formula, or the standard method-of-types bounds on a fixed alphabet, gives

$$\frac{1}{n} \log T_n \longrightarrow H(p) = \rho. \tag{3.4}$$

Now use separate exponential tilts on the two sides. Every vector in $V_n + V_n$ has coordinates in S and total coordinate sum at most $2L_n$. Since $0 < y < 1$,

$$|V_n + V_n| y^{2L_n} \le P_+(y)^n.$$

Because $2L_n \le n\mu$,

$$|V_n + V_n| \le e^{n\sigma}. \tag{3.5}$$

Likewise, if $z = a - b \in V_n - V_n$, then $\kappa(z_i) \le a_i + b_i$ coordinatewise. Hence

$$|V_n - V_n| x^{2L_n} \le P_-(x)^n,$$

and

$$|V_n - V_n| \le e^{n\rho}. \tag{3.6}$$

Fix $r > \rho$. Put $N_n = \lceil e^{rn} \rceil$, and use Bertrand's postulate to choose a prime

$$N_n < p_n < 2N_n.$$

For all sufficiently large n, equations (3.4)-(3.6) give

$$p_n > |V_n - V_n|, \qquad p_n > 2\max M, \qquad T_n \geq 2.$$

Apply Lemma 2.1 to V_n, W_n, p_n. It gives a finite $A_n \subset [0, p_n - 1]$ satisfying the strict GHR hypothesis and

$$|A_n - A_n| > T_n/2, \qquad |A_n + A_n| \leq 2e^{n\sigma},$$

$$2\max A_n + 1 = 2p_n - 1.$$

Therefore

$$\theta(A_n) > 1 + \frac{\log T_n - n\sigma - \log 4}{\log(2p_n - 1)}.$$

By (3.4), the numerator after the initial 1, divided by n, tends to $\rho - \sigma$. The denominator divided by n tends to r. For every number strictly below the right side of (3.2), one sufficiently large fixed n therefore supplies a finite GHR seed. Applying the GHR lemma to that single seed proves (3.2). Letting $r \downarrow \rho$ proves (3.1), and putting $r = \log R$ gives (3.3). □

What is new relative to the one-temperature theorem

The earlier cyclic-quotient proof used $y = x$, so its numerator collapsed to $\log P_-(x) - \log P_+(x)$. The sum upper bound does not require this: its Markov parameter can be optimized independently. The new theorem uses the difference tilt x to choose the high-entropy realizable type and the sum tilt y only to upper-bound all possible sums at the resulting budget.

For the gapless limiting alphabet $\mathbb{N}_0$, taking $x = \sqrt{2} - 1$ and $y = 1/2$ gives

$$\rho = 2\log(1 + \sqrt{2}), \qquad \sigma = 2\log 2,$$

and therefore

$$2 - \frac{\sigma}{\rho} = 2 - \frac{\log 2}{\log(1 + \sqrt{2})} = 1.213560298642605\ldots.$$

Thus the independently announced 1.21356 ... result of Logan Kleinwaks is the natural gapless baseline of the two-temperature theorem. The construction below crosses it by changing the inner alphabet, not by changing a cutoff in the old mask.

4. The structured alphabet $\langle 4, 5\rangle$

Let

$$H = \langle 4, 5\rangle.$$

It has conductor 12 and exactly six gaps:

$$\mathbb{N}_0 \setminus H = \{1, 2, 3, 6, 7, 11\}.$$

Take

$$M = H \cap [0, 80].$$

Exact reconstruction gives

$$|M| = 75, \qquad |M + M| = 155, \qquad |M - M| = 161.$$

In fact,

$$M + M = \{0, 4, 5, 8, 9, 10\} \cup [12, 160],$$

while

$$M - M = [-80, 80].$$

For every positive $d \in H \cap [1, 80]$, the witness $(d, 0)$ gives $\kappa(d) = d$. The six missing positive differences are recovered at the following exact costs:

d	1	2	3	6	7	11
$\kappa(d)$	9	18	13	14	17	19.

Thus the semigroup deletes six low-cost sum symbols but retains every difference symbol, paying a moderate witness-cost penalty only on those six gaps. That trade is exactly what improves the entropy ratio.

The partition polynomials can be written explicitly as

$$P_+(y) = 1 + y^4 + y^5 + y^8 + y^9 + y^{10} + \sum_{s=12}^{160} y^s,$$

and

$$P_-(x) = 1 + 2\left(\sum_{d \in H \cap [1,80]} x^d + x^9 + x^{18} + x^{13} + x^{14} + x^{17} + x^{19}\right).$$

5. Exact numerical certificate

Use

$$x = \frac{8149}{10000}, \qquad y = \frac{42571}{50000}, \qquad R = \frac{249293}{10000}.$$

The exact verifier reconstructs the polynomials and proves by integer cross-multiplication that

$$5.214722552042 < P_-(x) < 5.214722552043,$$

$$3.661102186601 < P_+(y) < 3.661102186602,$$

and

$$7.643532314898 < \mu < 7.643532314899.$$

It then uses rational atanh-series enclosures for every logarithm and proves

$$\log R - \rho > 0.00000427721454043878725250926 06, \tag{5.1}$$

and

$$\rho - \sigma - \frac{10709}{50000}\log R > 0.00000288188933267297001227382 47. \tag{5.2}$$

Equation (5.1) is the full-difference-rate condition in Theorem 3.1. Equation (5.2) gives

$$1 + \frac{\rho - \sigma}{\log R} > 1 + \frac{10709}{50000} = 1.21418.$$

Hence

$$\boxed{C_{3a} > 1.21418}.$$

Display-only values reconstructed from the exact integers are

$$\begin{aligned} P_-(x) &= 5.21472255204231355326778335618\ldots, \\ P_+(y) &= 3.66110218660138413370190644948\ldots, \\ \mu &= 7.64353231489850089011819946550\ldots, \\ \rho &= 3.21603954130634406125710246429\ldots, \\ \sigma &= 2.52722439436574711487654603261\ldots, \\ \log R &= 3.21604381852111829907290215884\ldots. \end{aligned}$$

6. The Lean-verified finite GHR seed

The formal proof uses one finite depth

$$N = 10^{13}, \qquad x = \frac{8149}{10000}, \qquad y = \frac{42571}{50000}.$$

Let $D = M - M = [-80, 80]$ and $Z = P_-(x)$. For $d > 0$, set

$$n_d = n_{-d} = \left\lfloor \frac{Nx^{\kappa(d)}}{Z} \right\rfloor, \qquad n_0 = N - 2\sum_{d=1}^{80} n_d, \qquad L = \sum_{d=1}^{80} n_d \kappa(d).$$

The exact finite checks in `formalization/lean/C3a/CertificateData.lean` prove that all counts are positive, their sum is N, and

$$n_0 = 1917647564302, \qquad 2L = 76435323145630.$$

Write $\mathcal{V}$ for the vector set, reserving V below for the integer seed after dilation:

$$\mathcal{V} = \left\{ a \in M^N : \sum_i a_i \le L \right\}.$$

Let $W \subset D^N$ be the type class with counts (n_d). Symmetry makes the two minimum-witness budgets equal to L, so $W \subset \mathcal{V} - \mathcal{V}$, and exact finite type counting gives

$$|W| \prod_{d \in D} n_d! = N!.$$

These statements and the independent tilted upper bounds

$$|\mathcal{V} + \mathcal{V}| y^{2L} \le P_+(y)^N, \qquad |\mathcal{V} - \mathcal{V}| x^{2L} \le Z^N$$

are proved in `C3a/VectorWords.lean`. No limit is taken.

The common envelope

$$n_d \le \frac{(N + 1000) x^{\kappa(d)}}{Z} \quad (d \in D)$$

covers the floors and the extra zero symbols. Effective Stirling bounds in `C3a/TypeEntropy.lean` imply

$$\log |W| \ge N \log Z - 2L \log x - 1000 - 161 \left(\frac{\log N}{2} + 1 \right).$$

The rational logarithm bounds in `C3a/LogNumerics.lean` establish the following conservative integer bounds, assembled in `C3a/Main.lean`:

$$\begin{aligned} \log |W| &\ge a = 32160395000000, \\ \log |\mathcal{V} + \mathcal{V}| &\le b = 25272245000000, \\ \log |\mathcal{V} - \mathcal{V}| &\le r = 32160396000000. \end{aligned}$$

Put $m = |\mathcal{V} - \mathcal{V}| + 161$. Mathlib's proved Bertrand postulate supplies a prime

$$m < p \le 2m.$$

In particular, $p > 160$ and $p > |\mathcal{V} - \mathcal{V}|$. The finite collision-counting proof in `C3a/CyclicCompression.lean`, together with `C3a/IntegerCompression.lean`, gives a linear map $\varphi : \mathbb{Z}^N \to \mathbb{Z}/p\mathbb{Z}$ such that the least-nonnegative representative set $A \subset [0, p-1]$ satisfies

$$0 \in A, \qquad \max A > 0, \qquad |A - A| > |W|/2, \qquad |A + A| \le 2|\mathcal{V} + \mathcal{V}|.$$

There is no rescaling to force -1 into the quotient image. Instead, use the integer seed $V = 2A = \{2a : a \in A\}$. Dilation preserves sumset and difference-set cardinalities and supplies

$$|V - V| < 2 \max V + 1 = 4 \max A + 1.$$

Moreover, since $|\mathcal{V} - \mathcal{V}| \ge 1$,

$$4 \max A + 1 \le 1296 |\mathcal{V} - \mathcal{V}|, \qquad \log(4 \max A + 1) \le r + \log 1296 < r + 8.$$

Logan Kleinwaks's proved GHR dilation principle applies to this one finite seed. With $\alpha = 10709/50000$, the final comparison is

$$\alpha(r+8)+b+2<a, \tag{6.1}$$

whose exact positive slack is 36384716.28656. Since $\log 4 < 2$, this implies that the GHR exponent of V exceeds $1+\alpha$. The closed Lean theorem is

```
C3a.c3a_gt_121418 : (60709 / 50000 : Real) < MaskedDigit.C3a
```

Here `MaskedDigit.C3a` is exactly Kleinwaks's definition in `RequestProject/GHRConstant.lean`. His admissibility predicate requires the construction for every $K > 1$, so its lower bound also implies the bound under the definition in Tao's problem 3a.

The prime, coefficient vector, and finite sets are specified by existence proofs, not materialized. The rate $R = 249293/10000$ belongs to the limiting comparison in §5; it is not used to choose the prime in this finite proof.

7. Verification boundary and attribution

The end-to-end Lean proof of the strict finite bound is `formalization/lean/C3a/Main.lean`. It includes the finite type argument, weighted counts, quotient existence, integer lifts, and final inequality. Bertrand's postulate is proved in Mathlib; the GHR finite-set and dilation principles are proved in the reused files `RequestProject/GHRDigits.lean` and `RequestProject/GHRConstant.lean`. They are not assumed axioms. The general limiting Theorem 3.1 is not formalized here and is not needed by the finite theorem.

The submitted proof pins Lean 4.35.0-rc2 and the Mathlib commit

```
065356127b1dc0016f66b7283ce0ce2c4055aa55
```

in `formalization/lean/lean-toolchain` and `lake-manifest.json`. The proof was first built and kernel-checked with Lean 4.28.0 and Mathlib v4.28.0 (commit `8f9d9cff6bd728b17a24e163c9402775d9e6a365`), then ported to Lean's module system for registration.

Run the axiom audit with `lake env lean AxiomAudit.lean` from `formalization/lean/`. The main theorem depends only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. All 17 original native finite checks now have kernel-checked proofs, using interval arguments, exact rational identities, `decide +kernel`, and `norm_num`. Neither `Lean.ofReduceBool` nor `Lean.trustCompiler` occurs in the theorem's axiom list.

The verbatim audit is in `formalization/lean/evidence/AXIOMS.txt`, and `evidence/KERNEL_CHECKS.md` records the replacements and timings. The earlier native version remains available at commit `5bbc2a5eb4d01eb3eb24fb909f819a8d220653f8`. Neither an independent agent review nor a local build is a claim of journal refereeing or maintainer acceptance.

Logan Kleinwaks's Apache-2.0 code is reused from `kleinwaks/masked-digit-sum-difference-bound`, commit `352ab86effecb7efce415507a8c781ec158b86e0`. The ten files under `RequestProject/` are unchanged apart from attribution headers. They provide the GHR definition and principles, mask infrastructure, type counting, Stirling estimates, and logarithm bounds. The new vector-word proof adapts his weighted-count and type-word arguments. The license and detailed attribution are in `formalization/lean/`.

The original Python and C++/GMP verifiers remain independent exact checks of the alphabet, costs, partition sums, and the §5 inequalities. They also check an older, different exact-Gibbs finite instance: its depth has 2129 bits and its prime is chosen above $\lfloor R^N \rfloor$, followed by rescaling. Accordingly, `certificate/N.txt`, `certificate/L.txt`, their hashes, and those programs' finite-size output describe that legacy instance, not the $N = 10^{13}$ Lean instance in §6. Those artifacts are preserved unchanged for reproducibility; they are not inputs or axioms of the Lean proof.

8. Reproduction

Primary finite proof, from `formalization/lean/`:

```
lake exe cache get
lake build
lake env lean AxiomAudit.lean
```

Independent numerical and legacy-instance checks, using Python's standard library only:

```
python3 verification/verify_c3a_two_temperature_121418.py --emit certificate
```

Independent C++/GMP:

```
g++ -O2 -std=c++20 verification/verify_c3a_two_temperature_121418_gmp.cpp \
  -lgmpxx -lgmp -o verify_gmp
./verify_gmp --emit certificate_gmp
```

Then compare the five discrete certificate files byte for byte.

9. Acknowledgements and AI-use disclosure

I thank Logan Kleinwaks for reviewing an earlier `1.2060` candidate, identifying minor statement and exposition issues that are corrected here, and publicly sharing his independent `1.21356` construction. I also thank the contributors to the Optimization Constants in Mathematics project for maintaining the problem record and its reproducibility standards.

Carson Olafson used ChatGPT extensively to explore constructions, formulate the two-temperature argument, draft and revise this note, generate the verification programs, and audit the resulting package. Carson selected the research direction, ran and reviewed the verification outputs, and is the human contributor responsible for this submission. The finite bound in §6 is formally verified; this note is not externally refereed. Independent review of the general Theorem 3.1 is expressly invited.

References

1. K. Gyarmati, F. Hennecart, and I. Z. Ruzsa, *Sums and differences of finite sets*, Functiones et Approximatio Commentarii Mathematici **37** (2007), 175-186, DOI 10.7169/facm/1229618749.
2. L. Kleinwaks, review of an earlier cyclic-quotient construction and announcement of an independent `1.21356` bound, optimizationproblems PR #134 comment, August 15, 2026.
3. L. Kleinwaks, *Improve C_3a lower bound via Lean-formalized controlled-carry semigroup mask-digit construction*, optimizationproblems PR #146, August 14, 2026.